

7. (a) In the table below, statements are given which describe different particles or interactions.

Complete the table by naming the particle or interaction described.

[4]

Description of particle or type of interaction	Name of particle or interaction
The quark combination of this particle is uud .	
The electron and electron neutrino are examples of this group of particles.	
Antibaryons are a combination of three of these.	
Neutrino involvement and quark flavour changes are exclusive to this type of interaction.	

- (b) An antiparticle has a quark composition of $\bar{u}\bar{d}\bar{d}$. Determine its charge and identify the particle. Show your working clearly.

[2]

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- (c) (i) Consider the following hypothetical interaction.



The reaction is **not possible** because it does not obey one or more of the conservation laws. By considering baryon number, lepton number and charge, show which law(s) are obeyed and which are not.

[3]

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- (ii) Jon suggests that replacing the electron neutrino (ν_e) with a pi-zero pion (π^0) would allow the reaction to occur. Explain whether or not he is correct.

[1]

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